

Sampling and Variability

Project: Verified Study Buddy

1. Sampling and Variability

Teach the learning cycle

Primary target: Sample from a distribution using explicit cumulative intervals

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Tokens and Context

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

Look for: Account for fixed text, evidence and reserved output under a toy budget

Reference example, not the only acceptable wording: Teacher explanation: $100 - 20 - 10 - 20 = 50$ supplied token units.

Predictive and Generative AI

Without opening the old lesson, explain one key idea from Predictive and Generative AI. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative prediction from generation by inspecting the requested output

Reference example, not the only acceptable wording: Teacher explanation: Message labels are discriminative prediction; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself. One application can route a request, retrieve evidence and generate a response.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Prompt A gives three identical wrong answers. Prompt B gives two correct answers and one wrong answer. Compare consistency and correctness without declaring either generally superior.

Concept to explain

Decoding choices trade consistency, diversity, and risk.

Term · Meaning

Term: decoding

Meaning: Selecting output tokens from a model's scores using a specified procedure.

Term: variability

Meaning: Differences observed across repeated runs, samples or conditions.

Term: temperature intuition

Meaning: A rough understanding of how temperature rescales token scores: higher values commonly spread choices more, without guaranteeing creativity or truth.

Term: seeded test

Meaning: A test containing a deliberately planted, known condition or error to check whether it is handled.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Use two 45-minute sessions. All model outputs and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed. Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

No. Later conditional probabilities can change path rankings, as the two-step counterexample shows.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

Apply the connected idea: A small attention calculation

Use two 45-minute sessions. All model outputs and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed. Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

Normalise the permitted weights by their remaining total; handle an empty set explicitly.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

End goal for the pictorial case

By the end, I can calculate a toy temperature change and trace a sampling draw.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Low variability favors high-weight continuations repeatedly.
2. Higher variability gives lower-weight options more chances.
3. Diversity may help brainstorming but can increase off-task or unsupported detail.
4. Choose settings by task, then evaluate repeated runs—not one favorite output.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

One prompt produces three different valid quiz questions.

Does variability mean failure?

Exact explanation to model

No. Sampling can vary outputs. Score correctness, relevance and format over repeated trials; consistency differs from quality.

Explain the added worked example

Start with probabilities red 0.6, blue 0.3, green 0.1 in that fixed order. Greedy chooses red. Sampling uses a draw u with $0 \leq u < 1$ and cumulative intervals: red $[0,0.6)$, blue $[0.6,0.9)$, green $[0.9,1)$. Left endpoints are included, right endpoints excluded. Thus $u=0.6$ selects blue and $u=0.9$ selects green. The draw 1 is outside this contract.

Sampling does not mean choosing all tokens equally. Across many independent draws the frequencies may approach the specified probabilities, but a small sample need not match them exactly. Fixed recorded draws allow a reproducible classroom trace. A random seed and generator version help replay a real

sampling program, not establish the truth of outputs.

For strictly positive probabilities p_i and temperature $T > 0$, define $q_i = p_i^{1/T} / \sum_j p_j^{1/T}$. This classroom formula is equivalent to scaling logits before a softmax when p came from those logits. We use it directly on the supplied table. At $T=1$ the distribution is unchanged. At $T=0.5$, square each probability: 0.36, 0.09, 0.01; total 0.46. The adjusted distribution is red 18/23, blue 9/46, green 1/46.

Key terms

Term · Meaning

Term: Sampling

Meaning: Selecting an alternative using a specified random procedure

Term: Greedy choice

Meaning: Selecting an alternative with largest next-token probability under a tie rule

Term: Temperature

Meaning: A positive parameter that changes the concentration of a distribution under a stated transformation

Term: Cumulative interval

Meaning: A range of draws assigned to an alternative using accumulated probability mass

Apply the connected idea: A small attention calculation

For one query q and keys k_i of width d , compute score $s_i = \text{dot}(q, k_i) / \sqrt{d}$. Then obtain weights from softmax and combine value vectors. Dot products may be negative; probabilities cannot. The softmax performs the conversion. This is a single-head classroom calculation, not a full language model.

Define softmax weights as $\exp(s_i - m) / \sum_j \exp(s_j - m)$, where m is the maximum permitted score. Subtracting the same maximum from every permitted score leaves the ratios unchanged and avoids unnecessary overflow from large positive scores. Natural exponent $\exp$ is introduced here; use a calculator or Python `math.exp` rather than estimating its meaning from the word exponential.

Take $q=[1]$ and $d=1$. The scale $\sqrt{1}$ is 1, so scores are 0 and $\ln(3)$. $\ln(3)$ is the number whose natural exponential is 3, approximately 1.098612. Exponentials have ratio 1:3, hence weights $1/4$ and $3/4$. Output is $(1/4)*2 + (3/4)*6 = 5$. Using maximum subtraction gives exponentials $1/3$ and 1 and the same normalised weights.

Guided: shifted scores $-\ln(3), 0$; exponentials $1/3, 1$; weights $.25, .75$; output $.25 \cdot 4 + .75 \cdot 8 = 7$. Exit: both coefficients are $.5$; opposite values cancel to zero.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The lower positive temperature concentrates this table on its largest entry. For $T=2$, use square roots before normalising; this flattens this unequal distribution. These changes leave the order of positive probabilities unchanged, so greedy still selects red. They change the sampling distribution, not the factual evidence behind a generated claim.

Our formula excludes $T=0$; dividing by zero is undefined. Some systems offer a zero-temperature setting as a separate greedy policy, but that is not a value to substitute into this formula. We also avoid claiming that any one setting removes all nondeterminism in an unspecified system.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sampling and Variability

A fictional two-token distribution illustrates variability in generation.

The facts we will use

At $T=1$, $A=0.8$ and $B=0.2$. For positive T use $p^{(1/T)}$, then normalise.

Use token order A, B and cumulative intervals $[0, \text{boundary}), [\text{boundary}, 1)$. Draw $u=0.85$.

Temperature-scaled toy probabilities

	T=1	T=0.5
A	0.8	0.9411764705882353
B	0.2	0.058823529411764705

Read a cell using both its row and column labels.

Sampling and Variability

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Original distribution

A boundary 0.8; $u=0.85$ selects B.



2. $T=0.5$ weights

Square probabilities: 0.64 and 0.04; total 0.68.



3. Normalise

$A=16/17 \approx 0.9412$; $B=1/17 \approx 0.0588$.



4. Same draw

0.85 is below $16/17$, so $T=0.5$ selects A.

CHANGE ONE THING

Set the draw to 0.97, keeping both distributions.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Original distribution

What happens and why: A boundary 0.8; $u=0.85$ selects B.

Step: 2. $T=0.5$ weights

What happens and why: Square probabilities: 0.64 and 0.04; total 0.68.

Step: 3. Normalise

What happens and why: $A=16/17 \approx 0.9412$; $B=1/17 \approx 0.0588$.

Step: 4. Same draw

What happens and why: 0.85 is below 16/17, so $T=0.5$ selects A.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Lower positive temperature concentrates this unequal distribution on its larger entry. It changes sampling probabilities, not evidence for whether A or B is true. Using the same draw isolates how the distribution changes the decision. This formula is a toy probability-space illustration corresponding to a temperature-scaled model distribution.

What this example does not establish

Two tokens and one draw do not predict every output; temperature alone does not solve hallucinations.

Change one thing in the pictured case

Change: Set the draw to 0.97, keeping both distributions.

Prediction: Both temperatures select B.

Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: One prompt produces three different valid quiz questions.

Write the question to explain: Does variability mean failure?

Reveal after predictions: No. Sampling can vary outputs. Score correctness, relevance and format over repeated trials; consistency differs from quality.

Keep for an independent attempt: Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: $T=1$ gives red,blue,green; $T=0.5$ gives red,red,blue. Three fixed draws check the procedure, not expected long-run proportions. Exit: concentration changed; the claim of impossible factual errors lacks source-grounded evaluation and cannot be established by temperature alone.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and distinguish a numerical claim from its interpretation.

Stage: 7 to 18 minutes

Teacher action: Read section 1 and complete W1; ask for a reason alongside each answer.

Stage: 18 to 32 minutes

Teacher action: Trace the worked case in section 2 and complete W2.

Stage: 32 to 45 minutes

Teacher action: Discuss the diagram and complete W3 in pairs; record support.

Stage: Session 2 start to 8 minutes

Teacher action: Complete the guided response before sharing the answer.

Stage: 8 to 20 minutes

Teacher action: Read the remaining sections and trace another constructed case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, requiring calculations and a bounded conclusion.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit response. Provide feedback and a fresh retry.

Teach the connected idea

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and distinguish a numerical claim from its interpretation.

Stage: 7 to 18 minutes

Teacher action: Read section 1 and complete W1; ask for a reason alongside each answer.

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Stage: 32 to 45 minutes

Teacher action: Discuss the diagram and complete W3 in pairs; record support.

Stage: Session 2 start to 8 minutes

Teacher action: Complete the guided response before sharing the answer.

Stage: 8 to 20 minutes

Teacher action: Read the remaining sections and trace another constructed case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, requiring calculations and a bounded conclusion.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Completion task: At $T=0.5$ compute squared weights, total and normalised probabilities. Check the sum.

Teacher check: Teacher explanation: Weights 0.36,0.09,0.01 sum 0.46. Probabilities 18/23,9/46,1/46 sum 1.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Sampling and temperature. Proposed explanation: “Sampling means equal chances”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Intervals have widths equal to the probabilities; unequal widths give unequal chances.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: A small attention calculation

Target: Compute scaled query-key scores and softmax weights

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Use values [2],[6] for the preceding weights. Compute output and output when the second key is excluded.

Teacher check: Teacher explanation: Full output 5; with B excluded, weight $A=1$ and output 2.

Individual check: For A small attention calculation, use its worked case to evaluate this proposed explanation: “Scores already are probabilities”. Explain the deciding step.

Expected correction: Scores may be negative or fail to sum to one; softmax supplies normalised nonnegative weights.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Does $T=0.5$ make B impossible, and does choosing A guarantee factual accuracy?

Reasoned teacher answer

No. B still has probability $1/17$, and the table supplies no truth labels for either token.

Changed-case reasoning: Both temperatures select B. 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Run each setting five times on one fixed prompt, score consistency, usefulness, and unsupported detail, then justify a task-specific choice.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Sampling means equal chances: Intervals have widths equal to the probabilities; unequal widths give unequal chances.

$T=0$ can be inserted into $p^{(1/T)}$: The formula requires $T>0$; greedy is a separate policy.

Lower temperature verifies claims: A probability transformation does not add factual evidence.

Check these misconceptions

Scores already are probabilities: Scores may be negative or fail to sum to one; softmax supplies normalised nonnegative weights.

Value width must equal key width: Values share a common output width, which can differ from query/key width.

Zero output means no source contributed: Nonzero weighted contributions may cancel.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

No. Sampling can vary outputs. Score correctness, relevance and format over repeated trials; consistency differs from quality.

Independent question

Prompt A is run three times and returns three different stories. Does this alone prove the prompt changed? Describe a fair comparison of two prompts.

Expected independent answer

Sampling can produce variation with an unchanged prompt. Run both prompts on the same tasks, record settings, repeat trials and judge with the same rubric.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Any difference between outputs proves a difference between prompts.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

A is consistent but 0/3 correct; B is less consistent but 2/3 correct on these trials. Repeat on predefined varied cases and keep conditions fixed. Identical output is not evidence of quality.

Explanations for the additional workbook examples

W1 Trace boundaries

At $T=1$ select tokens for $u=0,0.6,0.9$ and 0.999 . Explain the handling of $u=1$.

Teacher explanation: Selections red,blue,green,green. The draw 1 violates the half-open $[0,1)$ input contract and must be rejected.

W2 Compute the adjusted table

At $T=0.5$ compute squared weights, total and normalised probabilities. Check the sum.

Teacher explanation: Weights 0.36,0.09,0.01 sum 0.46. Probabilities $18/23, 9/46, 1/46$ sum 1.

Under the lesson's supplied temperature procedure, $T=0.5$ means exponent $1/T=2$. Square the weights to obtain 0.36, 0.09 and 0.01; their total is 0.46. Divide each by 0.46 to obtain $18/23$, $9/46$ and $1/46$. Converting $18/23$ to $36/46$ shows that the sum is $(36+9+1)/46 = 1$.

W3 Compare a fixed draw

Trace $u=0.7$ at $T=1$ and $T=0.5$. Does the different output demonstrate improved factual accuracy?

Teacher explanation: $T=1$ selects blue; $T=0.5$ selects red. This follows changed intervals and supplies no factual correctness evidence.

W4 Sample a new table independently

Probabilities $A=0.5, B=0.3, C=0.2$ in that order. Compute $T=0.5$ probabilities and cumulative boundaries. Trace $u=0.5$ and $u=0.9$ under $T=1$ and $T=0.5$. Explain the limit of the comparison.

Teacher explanation: Squared weights 0.25,0.09,0.04 total 0.38. Probabilities $25/38, 9/38, 4/38$; boundaries $25/38$ and $34/38$. $T=1$: 0.5 selects B, 0.9 selects C. $T=0.5$: 0.5 selects A, 0.9 selects C because $0.9 > 34/38$. The trace shows distribution effects, not factual accuracy.

W5 Specify a valid experiment

A student changes the prompt and model as well as temperature, then attributes a better answer to temperature. Propose a fair comparison and reject any unsupported universal claim.

Teacher explanation: Fix prefix/prompt, model/version, candidate distribution and sampling policy while varying temperature. Record draws and evaluate stated task criteria across cases. The mixed comparison

cannot isolate temperature; even a controlled result cannot prove all future outputs correct.

A small attention calculation — W1 Compute scores and weights

For $q=[1]$, keys $[0],[ln(3)]$ and $d=1$, compute scores and softmax weights.

Teacher explanation: Scores $0, ln(3)$; exponential ratio $1:3$ gives weights $1/4, 3/4$.

The dot products are $1*0=0$ and $1*ln(3)=ln(3)$. Division by $\sqrt{d}=\sqrt{1}=1$ leaves them unchanged. Exponentiation gives 1 and 3, with total 4. Normalising gives weights $1/4$ and $3/4$. These are compatibility weights under the stated calculation, not truth scores.

A small attention calculation — W2 Combine and mask

Use values $[2],[6]$ for the preceding weights. Compute output and output when the second key is excluded.

Teacher explanation: Full output 5; with B excluded, weight $A=1$ and output 2.

A small attention calculation — W3 Trace equal scores

For $q=[1,1]$, keys $[1,0],[0,1]$, values $[2,0],[0,4]$, compute scaled scores, weights and output.

Teacher explanation: Both scores $1/\sqrt{2}$; weights $.5, .5$; output $[1,2]$.

The query has dot product 1 with each key. Both scores are therefore $1/\sqrt{2}$. Equal scores give equal softmax weights, 0.5 each. Combine the values coordinate by coordinate: $0.5*[2,0] + 0.5*[0,4] = [1,2]$.

A small attention calculation — W4 Calculate new attention independently

$q=[1]$, keys $[0],[ln(2)],[ln(3)]$, values $[0],[3],[8]$, width 1. Compute scores, weights and output. Then mask the third item and recompute. Explain whether output establishes a factual claim.

Teacher explanation: Scores $0, ln(2), ln(3)$; weights $1/6, 2/6, 3/6$. Output $0+1+4=5$. Mask third: weights $1/3, 2/3, 0$; output 2. Neither weighted output verifies a factual claim.

A small attention calculation — W5 Check stability and shape

Scores are 1000 and $1000+ln(3)$; values 4 and 8. Compute stable shifted weights and output. What if one key has width two and the query width one?

Teacher explanation: Shift to $-ln(3), 0$; weights $1/4, 3/4$ and output 7. Mismatched key/query widths must be rejected rather than silently truncating.

Assess independent transfer

Selected case: Probabilities $A=0.5, B=0.3, C=0.2$ in that order. Compute $T=0.5$ probabilities and cumulative boundaries. Trace $u=0.5$ and $u=0.9$ under $T=1$ and $T=0.5$. Explain the limit of the comparison.

Selected case answer

Teacher explanation: Squared weights $0.25, 0.09, 0.04$ total 0.38. Probabilities $25/38, 9/38, 4/38$; boundaries

25/38 and 34/38. T=1: 0.5 selects B, 0.9 selects C. T=0.5: 0.5 selects A, 0.9 selects C because $0.9 > 34/38$. The trace shows distribution effects, not factual accuracy.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Verified Study Buddy

Use low variability for structured study answers and allow more diversity only for labeled brainstorming.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, input, output and evidence. Stretch: introduce

an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 5: Prompt as a Contract. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it. Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires probabilities 25/38, 9/38, 4/38, correct boundaries and B/C versus A/C. Fresh retry: $A=0.4, B=0.4, C=0.2$; at $T=0.5$ probabilities 4/9, 4/9, 1/9. Draws 0.4 and 0.85 select B, C at $T=1$ but A, B at $T=0.5$. Greedy ties must use an explicit rule.

Week 5 connects token IDs to embedding vectors and introduces position information without treating IDs as semantic measurements.

Readiness and reassessment for A small attention calculation

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it. Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires weights 1/6, 1/3, 1/2 and outputs 5 then 2. Fresh retry: keys [0], [ln(3)], [ln(4)], values [2], [6], [0]. Weights 1/8, 3/8, 4/8; output 2.5. Exclude third: weights 1/4, 3/4, 0; output 5.

Plan the delayed return

Suggested return points in this level: ai-3/5, ai-3/7, ai-3/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Keep Study Buddy generation settings fixed during prompt comparisons and check evidence regardless of variability.

7. Prepare, teach and assess

Teaching target: Sample from a distribution using explicit cumulative intervals

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: At $T=0.5$ compute squared weights, total and normalised probabilities. Check the sum.

Reasoned practice answer: Teacher explanation: Weights 0.36, 0.09, 0.01 sum 0.46. Probabilities 18/23, 9/46, 1/46 sum 1.

Check every learner before discussion: Use the worked case for Sampling and temperature. Proposed explanation: “Sampling means equal chances”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Intervals have widths equal to the probabilities; unequal widths give unequal chances.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Set the draw to 0.97, keeping both distributions. Require a prediction and a reason before comparing. Expected result: Both temperatures select B. Reason: 0.97 exceeds both A boundaries, 0.8 and approximately 0.9412.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.